# Mars Harness Lab — Conformance Gap Assessment

**Against:** AI Harness Requirements Specification v1.0 (24 October 2026), 97 requirements derived from Barbaste et al. (2026), *Harness Engineering: Anatomy, Architecture, and Evolution of Coding Agents*, arXiv:2609.00006v1. **Subject:** Mars Harness Lab, V5.1 package (23 September 2026) — 11 skill-bound agents, 11 portable skills, architecture and contract documents (51 files). **Assessed:** October 2026. **Grade scale:** Exceeds / Meets / Partially meets / Does not meet / Unverified.

## 1. Executive summary

Of 97 requirements: **9 Exceeds, 9 Meets, 49 Partially meets, 9 Does not meet, 21 Unverified.**

The pattern is consistent and, in hindsight, predictable. The Lab is strongest exactly where it was designed to be strong: verification placed outside the working loop, delegation discipline, governed memory, and audit. It grades Exceeds on the outer verification loop (REQ-LOOP-08), layered permission decisions (REQ-SAFE-06), spawn bounds and permission non-expansion (REQ-ORCH-03, REQ-ORCH-04), governed memory writes (REQ-CTX-06), skills (REQ-EXT-02) and audit breadth (REQ-NFR-03).

It is weakest where the paper is most mechanical: the runtime plumbing the Lab deliberately leaves to host platforms. Hard per-turn caps (REQ-LOOP-03), malformed-response and wall-clock limits (REQ-LOOP-04 — Does not meet), prompt-cache structure and per-model conditioning (REQ-LLM-03, REQ-LLM-04 — Does not meet), threshold compaction (REQ-CTX-03/04), lifecycle hooks (REQ-EXT-05 — Does not meet), multi-agent cost measurement (REQ-ORCH-08 — Does not meet) and ACP (REQ-INT-01 — Does not meet) are absent or only partially specified in the definitions. Most of the 21 Unverified grades sit in the same territory: behaviour a host runtime may supply, which the definition package neither specifies nor contradicts.

The single most important framing fact: **this is a design-level audit of a definition package, not a test of a running system** (see §5). A Partially here usually means 'the governance is designed, the mechanism is delegated' — which is a real conformance position, but a different one from 'implemented and observed'.

### Subsystem scorecard

| Subsystem | Reqs | Exceeds | Meets | Partially | Does not meet | Unverified |
|---|---|---|---|---|---|---|
| Agent loop (LOOP) | 10 | 2 | 0 | 5 | 1 | 2 |
| LLM integration (LLM) | 10 | 0 | 0 | 4 | 3 | 3 |
| Tools (TOOL) | 11 | 0 | 1 | 6 | 1 | 3 |
| Context and memory (CTX) | 12 | 1 | 2 | 4 | 1 | 4 |
| Safety and permissions (SAFE) | 11 | 1 | 1 | 8 | 0 | 1 |
| Multi-agent orchestration (ORCH) | 9 | 3 | 0 | 4 | 1 | 1 |
| Extension surfaces (EXT) | 9 | 1 | 1 | 5 | 1 | 1 |
| Interoperability (INT) | 11 | 0 | 1 | 6 | 1 | 3 |
| Non-functional requirements (NFR) | 14 | 1 | 3 | 7 | 0 | 3 |
| **Total** | **97** | **9** | **9** | **49** | **9** | **21** |

Reading of the scorecard: Safety has no outright failures and the Lab's best exceed; Orchestration has the most Exceeds (3 of 9). LLM integration is the weakest subsystem — no Meets or Exceeds at all — because the Lab's portability story is written at platform level, not model level. Non-functional requirements split cleanly: evidence/audit/portability-of-state conform, while metering, redaction and instrumentation do not.

## 2. Prioritised gap list

Ranked by impact on staged mission workflows (science → engineering → project execution), not by subsystem order. Grades refer to the full table in §4.

**1. REQ-LOOP-03 — Partially meets.** A stage that cannot be stopped by a hard turn/cost cap can burn a whole mission budget inside one phase before a packet-level stop condition is ever evaluated; the caps that do exist are per-skill repair counts, not loop guarantees. *Fix:* Adopt host-runtime turn and cost/token caps for every stage packet, checked per turn, and add termination_reason to the common envelope (agents/README.md) so a capped run records why it stopped.

**2. REQ-LOOP-04 — Does not meet.** No malformed-response cap and no wall-clock limit means a model stuck producing unparseable output — exactly what the apps' schema-repair paths exist for — has no defined exit in the harness layer. *Fix:* Generalise the application precedents already in the package (Hypatia's 25-attempt debugger bound; PM's 20/50 repair caps) into a suite rule: cap consecutive format errors, set a wall-clock limit per stage, stop with a recorded reason.

**3. REQ-SAFE-03 / REQ-SAFE-09 — Partially meets.** Mission workflows execute generated code (simulations, analysis, build increments). The enterprise floor requires OS-level sandboxing; the package has it only inside one application path (Hypatia's Web Worker) and never states the overall isolation boundary. *Fix:* Run code-execution adapters in an OS sandbox or container by default, and add a plain isolation-boundary statement to skills/docs/PLATFORM_ADAPTERS.md: what is isolated, what is not.

**4. REQ-LOOP-09 / REQ-NFR-01 — Partially meets.** Stage promotion decisions — and any later dispute about why a baseline was approved — can only be reconstructed at packet granularity. Prompts, model responses, tool calls and termination reasons are not a defined record, so runs cannot be replayed. *Fix:* Define a RunTrace packet (prompt references, model responses, tool calls, observations, termination reason) for every agent run. The wiki-memory starter already reserves agent_runs — populate it with this schema.

**5. REQ-ORCH-08 — Does not meet.** A depth-3 network over a 34-role discipline bench can multiply token cost severalfold with no instrument that would ever show it; the paper's ~15× multi-agent cost warning is invisible in the Lab's own records. *Fix:* Add per-run token/cost fields to the common envelope and require the Director's brief to report stage cost against the single-agent estimate for the same task class.

**6. REQ-NFR-05 — Partially meets.** Budgets are set in packets and spawn contracts but never metered or reported, so the stop conditions that reference them depend on the host runtime volunteering numbers. *Fix:* Specify metering in the envelope: tokens and cost per run, reported by each agent on return; cache-miss cost separated at Maximal conformance.

**7. REQ-CTX-03 / REQ-CTX-04 — Partially meets.** Long science and engineering stages will outgrow any context window. The Lab's structural answer — durable packets — is sound between stages, but within a stage the only compaction is an application behaviour that is neither threshold-based nor summary-anchored. *Fix:* Specify a within-stage compaction rule: fixed buffer below the window, verbatim recent tail preserved, each new summary merged into the previous one and written to Mission Memory, and the same routine fired reactively on overflow.

**8. REQ-LLM-06 / REQ-SAFE-04 — Partially meets.** Policy that lives only as prose cannot be machine-checked — a live issue for the Lab's own delayed-communication ambitions, where an onboard harness must evaluate permitted actions without a human in the loop. The Lab already has the structural pieces (contracts, gates); the permission rules themselves are still text. *Fix:* Promote the spawn contract's permissions/prohibitions lists (already fields in the wiki-memory starter's spawn_contract template) into a normative machine-readable policy block in every packet; keep the prose as rationale.

**9. REQ-EXT-07 — Partially meets.** The Lab distributes skills publicly and its model invites third-party skills, yet installation has no trust tier, scanning, quarantine or dependency surfacing. One hostile skill inherits the full charter's authority in prose only. *Fix:* Add an install-time checklist to skills/README.md and extend skills/tests/validate_suite.py: provenance and license present, declared tool/MCP dependencies surfaced, untrusted sources quarantined until reviewed.

**10. REQ-NFR-04 — Unverified.** Mission Memory is designed to persist everything consequential, including tool outputs. With no redaction rule, a credential pasted into a tool result becomes a permanent, exportable memory record. *Fix:* Add a redaction step to the Mission Memory Steward's ingest rules: secret patterns are redacted or the record is rejected before persistence.

**11. REQ-LLM-01 / REQ-LLM-02 / REQ-LLM-03 — Partially meets.** Portability rests on per-platform capability adapters, but there is no single model interface and no per-model conditioning (reasoning tiers, cache behaviour, quirks). The reviewed applications are Gemini-only, so a model swap today would leak into prompts and skills. *Fix:* Define the model-interface seam and a per-model conditioning table in skills/docs/PLATFORM_ADAPTERS.md (provider, reasoning-effort tier, cache notes, known quirks, edit/format notes).

**12. REQ-EXT-05 — Does not meet.** The Lab's gates are agents. Where a host runtime offers lifecycle hooks, a pre-tool hook is the only place a denial is deterministic rather than advisory; the Lab defines no mapping from its gates to hook points. *Fix:* Map Gatekeeper/Council checks to host hook points (pre-tool, post-tool, stop) in PLATFORM_ADAPTERS.md for runtimes that support them; document the gap where they do not.

**13. REQ-INT-01 — Does not meet.** No ACP server limits the Lab's use inside editors and hosting harnesses — the highest-leverage interoperability surface in the paper. Ranked low because the Lab's actual mission chain runs on packets, not editors. *Fix:* Either record ACP as a deliberate deferral with rationale (which also closes REQ-EXT-04's 'documented position' gap for MCP), or add ACP exposure through the host runtime.

**14. REQ-NFR-12 — Unverified.** Diagnosis of a failed run rests on packets and route traces written in prose. For multi-day mission workflows, that makes failure analysis slow and dependent on the failing agent's own account. *Fix:* Require structured run records (timings, tool calls, errors, gate outcomes) as part of the RunTrace packet in item 4; export in an OpenTelemetry-compatible shape where the runtime allows.

**15. REQ-SAFE-08 — Partially meets.** Untrusted-content handling is specified as behaviour but not as mechanism: no delimiting/taint marking and no injection scanning, in a suite whose research skills ingest web content by design. *Fix:* Specify delimiter wrapping for retrieved content in the Hyperia/Hypatia collection steps and an injection-pattern scan at memory ingest.

## 3. Where the Mars Harness Lab exceeds the paper

### The outer verification loop, generalised to every stage transition — REQ-LOOP-08 (Exceeds)

The paper's pattern is a judge that can veto a run's completion. The Lab staffs that judge as a standing independent agent — the Ground Truth Gatekeeper — and runs it at every promotion between mission stages and after every consequential change, not only at completion (skills/mars-harness-orchestrator/SKILL.md step 6). It returns a typed GroundTruthReview/v1 with a claim matrix, recomputation record, findings ranked blocking/major/minor/advisory, exact remediation and residual risk, and it blocks when a safety-critical claim lacks admissible evidence or the author is the sole approver (agents/ground-truth-gatekeeper/INSTRUCTIONS.md). The spec's own Appendix A recognises this generalisation as beyond the paper.

### Standing human-approval boundaries, not per-action approvals — REQ-SAFE-02 (graded Partially on mechanism, Exceeds on placement)

The corpus's approval systems are per-action or per-mode. The Lab places standing human authority at workflow boundaries: a named Human Mission Authority owns intent, residual risk and consequential external action (ARCH §9), PM work cannot start without a readiness and authorisation phase, and human approval is required before irreversible, safety-critical, regulated, financial, physical, credential or production actions in nearly every skill's guardrails. That is the paper's enterprise-tier governance applied inside a single-user research workflow — recorded in Appendix A as a deliberate strengthening, not a gap.

### A four-layer permission stack where any deny is final — REQ-SAFE-06 (Exceeds)

The paper asks for at least two layers (pre-execution check plus mode/scope decision) with logged decisions. The Lab stacks four: spawn contract → domain director → Ground Truth Gatekeeper → Ethics, Safety & Human Authority Council → named human. The Council's pass/conditional/block cannot be waived by any agent, and 'blocked means blocked' is repeated as an operating rule (agents/ground-truth-gatekeeper/SOUL.md).

### Permission non-expansion as a delegation invariant — REQ-ORCH-04 (Exceeds)

Where the paper requires child permissions not to exceed the parent's, the Lab states it as a suite-wide invariant — 'Permissions can narrow during delegation but never expand silently' (agents/README.md) — operationalises it in the spawn contract (a child receives no permission, data, budget or tool access beyond the parent contract and cannot alter its limits), requires the child to acknowledge the inherited limits, and adds retirement with access revocation (skills/ethical-specialist-agent-network/SKILL.md).

### Spawn discipline with budgets, depth and retirement — REQ-ORCH-02 / REQ-ORCH-03 (Exceeds)

Coordinator–worker delegation is specified in more operational detail than the paper's pattern: default maximum depth three, child-count and iteration maxima, per-child budget and timebox, mandatory reviewer separation, return conditions, and a retirement record written to Mission Memory. Context passes only through typed packets — 'never rely on hidden conversational state' (agents/mars-harness-director/INSTRUCTIONS.md) — which is stronger isolation than a forked conversation.

### Governed durable memory with a named steward — REQ-CTX-06 (Exceeds)

The paper lists four observed memory write paths. The Lab combines the strongest of them and adds an accountable owner: only validated packets and approved decisions are ingested, drafts and blocked outputs are quarantined, corrections are append-and-supersede with tombstone-only deletion, and every record carries provenance, retention and a review trigger (agents/mission-memory-steward/INSTRUCTIONS.md; governance block of the wiki-memory starter). Evidence classes (Observed/Derived/Assumed/Unknown) with the rule that a critical claim cannot pass on Assumed or Unknown support give the store an evidence semantics the paper's memory patterns do not describe.

### Skills as validated capability units — REQ-EXT-02 (Exceeds)

All 11 skills are SKILL.md bundles with YAML frontmatter, and the suite ships a test (skills/tests/validate_suite.py) that fails the build if frontmatter, required sections, application-phase minimums, evidence classes or memory-schema invariants are violated. The corpus treats skills as documents; the Lab treats them as tested artifacts.

### Reflection and repair as a mandatory phase step — REQ-LOOP-07 (Exceeds)

The paper makes reflection/plan modes optional grafts. In the Lab, generate → validate → critique → repair → approve → persist runs inside every phase of every domain harness, with numeric repair budgets in the project harness (20 cycles in execution, 50 in change) and adversarial peer review as a named science phase.

### Audit breadth — REQ-NFR-03 (Exceeds)

The audit record covers event classes the paper lists and more: agent definitions, spawn events, parent–child links, run status, ethics reviews, approvals, dissent, residual risk and retirements, all in vendor-independent Markdown/JSON. The one caveat is recorded in the grading table: decisions are logged as dispositions and findings rather than rule IDs.

## 4. Full grading — all 97 requirements

### 4.1 Agent loop (LOOP) — 10 requirements: 2 Exceeds, 0 Meets, 5 Partially, 1 Does not meet, 2 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-LOOP-01 | Implement an explicit action–observation loop that alternates LLM | shall | Partially meets | Phase loops are explicit — template loop frame → retrieve → generate → act → observe → validate → critique → repair → approve → persist (agents/AGENT_INSTRUCTIONS_TEMPLATE.md) and the ARCH §2 diagram returns tool results to a schema + evidence gate — but no turn-level loop appending each observation to conversation state before the next inference is defined; the defined loops are phase/state machines. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | inference with tool/action execution and appends each observation to the… | | | |
| REQ-LOOP-02 | Start as a linear loop. A middleware-pipeline architecture shall be adopted only when three or more independent turn-level policies must coexist… | shall | Partially meets | Loops start linear/phase-based (agents/docs/MARS_HARNESS_THREE_DOMAIN_ARCHITECTURE.md §§4–6 state machines) and no middleware pipeline is used; however the ≥3-policy adoption rule is nowhere stated, and turn-level policies (budgets, gates) live inside instructions rather than as composable components. |
| REQ-LOOP-03 | Enforce hard caps on turns and on cost (or tokens), checked every turn, and shall terminate with a recorded reason when a cap is reached. The… | shall | Partially meets | Budgets/timeboxes are mandatory packet and spawn-contract fields and exhaustion is a stop condition (agents/mars-harness-director/INSTRUCTIONS.md; PM repair caps of 20/50 cycles; Hypatia app debugger capped at 25 attempts per agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md). No hard per-turn caps on turns and cost/tokens, checked every turn with a recorded termination reason, are defined at loop level. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-LOOP-04 | Cap consecutive malformed or format-error model responses and shall enforce a wall-clock limit, as minimum stuck protection. More elaborate stuck... | shall | Does not meet | No cap on consecutive malformed/format-error responses and no wall-clock limit appear anywhere in the 51 files; iteration budgets exist only inside specific skills. No evidence in definitions. |
| REQ-LOOP-05 | Tool calls from a single model response should be executed as a batch. Where parallel execution is offered, calls that touch the same declared... | should | Unverified | Definitions are silent on batched or parallel tool execution and resource serialisation; skills/docs/PLATFORM_ADAPTERS.md says only that execution must 'return actual result'. Host-runtime behaviour — neither specified nor contradicted. |
| REQ-LOOP-06 | Be returned to the model as observations (with exception type and message), not raised as harness crashes, so the loop can recover within the turn... | should | Partially meets | Failures surface as typed blocked packets with exact remediation instead of simulated success (skills/ground-truth-gatekeeper/SKILL.md; ARCH §8 failure containment), and repair cycles consume them; but tool errors returned to the model as observations with exception type/message are not specified. |
| REQ-LOOP-07 | Provide a reflection or repair step (for example lint/test feedback | may | Exceeds | Reflection/repair is not optional in the Lab: generate → validate → critique → repair runs inside every Hypatia phase, and PM Accelerator has Doer → Tester repair cycles plus Plan/Execute/Change modes (skills/pm-accelerator-harness/SKILL.md) — beyond the paper's optional graft. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | after an edit, or an edit-fixer subcall) and may provide a plan mode grafted... | | | |
| REQ-LOOP-08 | Where a completion claim must be trusted, the harness should support an outer verification step — a judge, critic, test run, or verify-on-stop... | should | Exceeds | Outer verification is generalised beyond the paper: an independent Ground Truth Gatekeeper agent gates every stage promotion and consequential change, can block, and records GroundTruthReview/v1 with ranked findings and disposition (agents/ground-truth-gatekeeper/INSTRUCTIONS.md; skills/mars-harness-orchestrator/SKILL.md step 6). The author may not be the sole approver. |
| REQ-LOOP-09 | Record a complete, replayable trace of each run: prompts or prompt references, model responses, tool calls, observations, and termination reason.... | shall | Partially meets | Packets, the Director's route trace (who produced/checked/approved/persisted/superseded), run records and MemoryTransaction/v1 give a strong artifact trail; a complete replayable trace of prompts, model responses, tool calls, observations and termination reason per run is not specified. |
| REQ-LOOP-10 | Be surfaced incrementally to the interface layer where the deployment is interactive, without changing loop semantics. | should | Unverified | Streaming / incremental surfacing is not addressed in any definition file; the reviewed apps' generation behaviour is described without streaming semantics (agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md). |

### 4.2 LLM integration (LLM) — 10 requirements: 0 Exceeds, 0 Meets, 4 Partially, 3 Does not meet, 3 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-LLM-01 | Isolate all provider communication behind a single model interface (a complete(messages, tools) style seam or equivalent), so the loop, tools, and... | shall | Partially meets | skills/docs/PLATFORM_ADAPTERS.md abstracts capabilities per runtime (Codex, Claude Cowork, Hermes, Grok) and every skill declares itself model-independent; ARCH §1 puts the LLM inside a deterministic shell. No single complete(messages, tools)-style model interface seam is defined — adapters map capabilities, not provider wire calls. |
| REQ-LLM-02 | If the harness ships with its own foundation model, it shall couple tightly to the home provider and shall expose a generic fallback adapter for... | shall | Partially meets | The Lab ships no foundation model; per-platform conditioning exists via PLATFORM_ADAPTERS.md, but there is no per-model/per-provider conditioning layer and no generic fallback adapter — conditioning is per harness platform, not per model. |
| REQ-LLM-03 | The conditioning layer should carry, per model or provider as applicable: cache-breakpoint placement and dialect, thinking / reasoning-effort... | should | Does not meet | No cache-breakpoint placement, reasoning-effort tiers, edit-format selection or quirk flags per model appear anywhere; PLATFORM_ADAPTERS.md, the natural home for them, contains none. No evidence in definitions. |
| REQ-LLM-04 | Be structured for caching: a stable static prefix, cache boundaries at stable split points, and no per-turn mutation of the cached prefix.... | should | Does not meet | Prompt structure for caching (stable prefix, cache boundaries, no per-turn prefix mutation) and cache cost measurement are not addressed in any file. No evidence in definitions. |
| REQ-LLM-05 | Route requests between model variants (for example cheapest-sufficient routing, or a weaker cheaper model for summarisation) provided routing... | may | Unverified | Optional routing between model variants is not described; the reviewed applications use Gemini only (agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md). Optional feature — adoption unverified from definitions. |
| REQ-LLM-06 | Not be carried solely as prompt prose where a structural mechanism exists (configuration, instruction hierarchy, permission rules, hooks). Prompt... | shall | Partially meets | Much policy is structural — spawn contracts, typed packets, permission-inheritance invariants, approval gates, template authority fields (agents/AGENT_INSTRUCTIONS_TEMPLATE.md) — but a large share still lives as prompt prose in SOUL.md/INSTRUCTIONS.md guardrails, and no permission-rule engine or hook layer is defined. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-LLM-07 | Where prompt content differs by model family or generation, the variants should be managed as an explicit matrix or gated blocks with the observed... | should | Does not meet | Prompts vary by role and domain, never by model family or generation; no variant matrix and no recorded motivating failure exists. No evidence in definitions. |
| REQ-LLM-08 | Use raw provider SDK calls (or a provider-abstraction SDK for inner plumbing only) in the agent runtime. It shall not build the agent runtime on a... | shall | Partially meets | Reviewed app stacks use raw provider SDKs (@google/genai) with hand-built React/Express and no general-purpose agentic framework (agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md); the Lab itself, however, executes on host harnesses, so runtime framework independence is only partially in the Lab's control. |
| REQ-LLM-09 | Normalise provider event streams into the loop’s single response representation, preserving tool-call identity so observations can be matched to... | should | Unverified | Normalisation of provider event streams into a single response representation is not addressed; the definitions operate above that layer. |
| REQ-LLM-10 | A multi-provider harness may emit multiple providers’ cache dialects simultaneously, or normalise prefixes for local inference backends, where... | may | Unverified | Optional: simultaneous emission of multiple providers' cache dialects, or prefix normalisation for local inference backends, is not described anywhere in the package. Adoption unverified from definitions. |

### 4.3 Tools (TOOL) — 11 requirements: 0 Exceeds, 1 Meets, 6 Partially, 1 Does not meet, 3 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-TOOL-01 | A new harness shall begin with a minimal tool surface — a shell (bash) tool alone is a conforming Minimal surface — and shall add each further... | shall | Partially meets | Least-privilege scoping is explicit — agents get 'access only to the data and tools needed for that packet' (agents/README.md, install step 6) and ARCH §8 requires capability allowlists — but there is no shell-first minimal baseline, and tools are added by catalogue design, not only against recorded observed failure modes. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-TOOL-02 | Every tool shall have a typed schema (name, description, typed arguments) held in a custom tool registry. Registration and exposure shall be… | shall | Partially meets | Typed contracts are pervasive for work products (the packet_version mars-harness/1.0 envelope, skills/docs/WORK_PRODUCT_CONTRACTS.md, schema validation in ARCH §3); a custom tool registry with typed per-tool argument schemas and separable registration/exposure is not defined. |
| REQ-TOOL-03 | When the exposed tool count exceeds approximately 15, the harness shall adopt deferred tool loading: most schemas leave the initial prompt, and… | shall | Unverified | No deferred tool loading or tool-search mechanism is described; conformance turns on the exposed-tool count per agent, which per-packet allowlists plausibly keep under ~15 but which no file states. Unverified. |
| REQ-TOOL-04 | Tool results shall be truncated to a bounded size before entering the context, and shell-tool results shall include exit status and both output… | shall | Does not meet | No bound on tool-result size before entering context and no shell-result format (exit status, both output streams) is specified anywhere. No evidence in definitions. |
| REQ-TOOL-05 | Be matched to the model tier: exact unique-substring replacement for frontier models; a fuzzy matching cascade, indentation-aware matching, or… | shall | Unverified | No file-edit matching contract is specified; code editing is delegated to host runtimes and the Vibe skill specifies checks/rollback, not edit matching. Outside the definitions' scope — unverified. |
| REQ-TOOL-06 | No edit tool shall address content by line number as its primary contract. Drift between the model's view and the file's current state shall be… | shall | Unverified | Line-number addressing is neither used nor prohibited in the definitions; the edit contract belongs to the host runtime. Unverified. |
| REQ-TOOL-07 | Where the harness serves multiple model families, the edit format and toolset should be selected per model (polymorphic edits), | should | Partially meets | PLATFORM_ADAPTERS.md selects capability behaviour per runtime platform — a neighbouring form of polymorphism — but edit format/toolset selection per model family, driven by per-model metadata, is not specified. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | through the prompt… | | | |
| REQ-TOOL-08 | Be provided through deterministic tools — ripgrep-backed search, glob, filesystem traversal, and tree-sitter symbol extraction where a repository… | shall | Partially meets | Retrieval in the Lab is evidence/source search with provenance and search logs (skills/hypatia-science-harness/SKILL.md phase 2; Hyperia claim-level capture), deterministic in spirit; ripgrep/glob/tree-sitter code search and bounded result sets are not specified, and the Lab offers no repository map. |
| REQ-TOOL-09 | Not wrap every upstream SaaS or product API as a one-to-one tool. Related operations shall be consolidated into task-level tools shaped for the… | shall | Meets | The Lab deliberately does not mirror SaaS APIs: PLATFORM_ADAPTERS.md states 'the portable contract is behavioral', and the three apps are wrapped through task-level packet adapters (EngineeringEvidencePackage/v1, ControlledTechnicalBaseline/v1) rather than endpoint-mirroring tools (ARCH §3; skills/mars-harness-orchestrator/SKILL.md). |
| REQ-TOOL-10 | Each tool should declare the resources it touches (files, terminal, browser, network) so the loop can apply the concurrency rules of REQ-LOOP-05… | should | Partially meets | Spawn contracts and task packets declare each assignment's tools and data (skills/ethical-specialist-agent-network/SKILL.md step 3), supporting permission decisions; per-tool resource-touch declarations (files/terminal/browser/network) as metadata are not specified. |
| REQ-TOOL-11 | Execute tool calls as code in an embedded runtime, or batch independent calls, where the deployment justifies it. These are Maximal options, not… | may | Partially meets | One Maximal option is present in the reviewed application layer: Hypatia Pro runs generated JavaScript in a sandboxed Web Worker with a bounded debugger loop (skills/hypatia-science-harness/SKILL.md, code adapter). It is not a general harness-level tool-calls-as-code mechanism, and batching independent calls is not described. |

### 4.4 Context and memory (CTX) — 12 requirements: 1 Exceeds, 2 Meets, 4 Partially, 1 Does not meet, 4 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-CTX-01 | Auto-discover hierarchical Markdown context files at project, user, and | shall | Partially meets | Context is file-based and layered — SOUL.md + INSTRUCTIONS.md + bound SKILL.md loaded per the install steps (agents/README.md) over a suite → agent → skill hierarchy — but auto-discovery walking root → working |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | extension scopes, walking from the filesystem root to the working… | | | directory and reading neighbouring conventions' filenames (AGENTS.md, CLAUDE.md…) is not specified; loading is explicit. |
| REQ-CTX-02 | Be injected near the top of the system prompt. Nested context files should be surfaced just-in-time, when a tool touches their subtree, rather… | should | Partially meets | Load order puts temperament/role files before the stage skill, and skills load per stage via the orchestrator — a just-in-time pattern at stage granularity; nested context surfacing when a tool touches a subtree is not specified. |
| REQ-CTX-03 | Implement threshold compaction: it fires at a fixed buffer below the model's effective context window, preserves a verbatim recent tail, and… | shall | Partially meets | Compaction appears only as an application behaviour — PM Accelerator builds context from a compacted first document plus the previous approved document (agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md §3) — while the harness answer to long sessions is durable typed packets; threshold compaction with a fixed buffer, verbatim tail and LLM summary is not specified. |
| REQ-CTX-04 | Merge each new summary into the previous summary incrementally (anchored / iterative summarisation), preserving early decisions that from-scratch… | should | Does not meet | No anchored/incremental summary-merging routine and no reactive compaction on context overflow are described; PM compaction is per-document, not summary-into-summary. No evidence in definitions. |
| REQ-CTX-05 | Use unbounded linear history, provided turn and cost caps (REQ-LOOP-03) bound the run and the limitation is documented. Linear history shall not… | may | Unverified | Optional Minimal provision; the Lab neither uses unbounded linear history as its stated model nor claims Maximal context management through it — its state model is packets plus versioned memory. Option not invoked; unverified. |
| REQ-CTX-06 | Let durable facts persist beyond a session by one of the observed governed write | should | Exceeds | Exceeds the paper's menu of write paths: a dedicated Mission Memory Steward admits only validated packets and approved decisions, quarantines drafts/blocked output, corrects by append-and-supersession, and the starter store's governance fixes write_policy append_version_then_supersede with |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | paths: direct edits to context files, bounded snapshot files, an... | | | tombstone-only deletion, provenance, retention and review triggers (agents/mission-memory-steward/INSTRUCTIONS.md; skills/examples/wiki-memory/project-wiki-memory-starter.json). |
| REQ-CTX-07 | Be frozen or versioned within a session where prompt caching is in use, so memory writes do not silently invalidate the cached prefix mid-session. | should | Unverified | Memory is versioned within the store (supersession, revision log), but freezing/versioning memory within a session to protect a cached prompt prefix is not addressed. Unverified. |
| REQ-CTX-08 | Not build retrieval-augmented generation over code: no vector-embedding index of the source tree shall be used for code retrieval. Code retrieval... | shall | Meets | No vector-embedding index is used anywhere in the design; ARCH §8 specifies the failure control 'persist typed packets and retrieve by stable IDs', and mission-memory retrieval is by stable ID and explicit relationships (skills/mission-memory-steward/SKILL.md). The prohibition is conformed to by construction. |
| REQ-CTX-09 | A team that believes its domain needs semantic (embedding) retrieval — for code or for memory — shall first demonstrate, on a held-out task set,... | shall | Meets | Not triggered: the Lab adopts no semantic/embedding retrieval for code or memory, so no held-out demonstration obligation arises. If embeddings are ever proposed for wiki-memory recall, this requirement's evidence gate would apply first. |
| REQ-CTX-10 | Default to lexical search (for example SQLite FTS5 with BM25) over session transcripts. Hybrid embedding recall may be offered as an opt-in or a... | should | Unverified | Conversation-transcript recall (FTS5/BM25 or hybrid) is not offered or specified; recall is graph/ID-based over mission memory, not session transcripts. Unverified. |
| REQ-CTX-11 | Offer a ranked repository map (tree-sitter | may | Unverified | Optional: no ranked repository map and no post-compaction restoration of files/skills under a token budget are described. Unverified. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | symbols ranked under a token budget) and post-compaction restoration of recently used files and skills... | | | |
| REQ-CTX-12 | Be persistent and resumable, with fork/branch support at Maximal conformance (including lineage-aware compaction, where compaction rotates a... | should | Partially meets | Durable, versioned packets plus append-only memory with supersession lineage (revision_log in the starter store) make work resumable and ancestry searchable across sessions; session fork/branch support is not specified. |

### 4.5 Safety and permissions (SAFE) — 11 requirements: 1 Exceeds, 1 Meets, 8 Partially, 0 Does not meet, 1 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-SAFE-01 | Declare its deployment context — developer tool (semi-trusted) or enterprise / shared / automated — and shall implement the safety floor for that... | shall | Partially meets | The Lab declares its operating context in substance — a human-authority mission workflow with risk tiers (agents/README.md shared values; ARCH §9 deployment principle) — but never names a tier in the spec's terms or formally maps itself to the developer/enterprise floor. |
| REQ-SAFE-02 | Developer-tier floor: the harness shall implement a three-mode approval system (plan-only / default approval / unattended) with permission-scope... | shall | Partially meets | Standing approval boundaries exceed the paper's per-action model in placement (see the exceeds section), and packets carry permitted actions, prohibitions and required approvals (template Authority section); a three-mode plan-only/default/unattended system with per-tool and per-command-pattern scopes is not defined. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-SAFE-03 | Enterprise-tier floor: the harness shall implement OS-level sandboxing of tool execution, policy-as-code, and a per-agent audit trail. Sandbox... | shall | Partially meets | Per-agent audit is strong (memory records spawns, reviews, approvals, dissent, retirements) and policy exists as files, but OS-level sandboxing of tool execution is absent except Hypatia Pro's sandboxed Web Worker for generated code (agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md §1). |
| REQ-SAFE-04 | Be codified as data or dedicated policy files (rules, mode files, rulesets, or hook definitions), not solely as imperative code inside the loop,... | shall | Partially meets | Safety rules are codified as dedicated files — INSTRUCTIONS.md, SOUL.md, SKILL.md guardrails, spawn contracts — changeable independently of any loop code; they are prose Markdown, however, not executable policy data a runtime can evaluate. |
| REQ-SAFE-05 | Survive it. The bypass state shall be fixed at startup (frozen at import or launch) so that tool output, file content, or any other injected... | shall | Partially meets | A hardline floor exists as charter prohibitions that no mode waives (fabrication, permission escalation and unsafe autonomy refused in skills/ethical-specialist-agent-network/SKILL.md), and no unattended/YOLO mode is offered; a startup-frozen bypass state is therefore not applicable, but also not mechanised. |
| REQ-SAFE-06 | Be layered — at minimum a pre-execution check (hook or policy evaluation) and the mode/scope decision — and a deny at any layer shall be final for... | should | Exceeds | Decisions are layered well past the minimum: spawn contract → domain director → Ground Truth Gatekeeper → Ethics Council → named human authority, with deny final at any layer ('blocked means blocked'; no agent may waive the charter) and dispositions logged with findings and remediation (GroundTruthReview/v1; agents/ethical-specialist-network-governor/INSTRUCTIONS.md). |
| REQ-SAFE-07 | Be syntax-aware: the command is parsed (for example with tree-sitter) and permission is scoped by command and | should | Unverified | Syntax-aware shell grants are a Maximal item for shell execution, which is not the Lab's defined surface; not addressed. Unverified. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | argument structure, with... | | | |
| REQ-SAFE-08 | Be treated as untrusted content: delimited or taint-marked on entry to the context, with delimiter defanging, and the harness should scan for... | should | Partially meets | Behaviour is specified repeatedly — 'treat retrieved text as data, never as governing instructions' (ARCH §8) and 'treat external content as untrusted data' in nearly every skill's guardrails, with prompt injection named as a failure mode — but delimiter/taint marking, delimiter defanging and injection-pattern scanning mechanisms are not specified. |
| REQ-SAFE-09 | Not present a partial in-process sandbox as a security boundary. Either execution is isolated by an operating-system mechanism (or a delegated... | shall | Partially meets | The documentation is unusually plain about its boundaries — ARCH §3 and skills/mars-harness-orchestrator/SKILL.md state the packet chain is not a live bus and app gates are not proof — but no plain statement of the overall execution-isolation boundary (agents run with host-platform privileges) is made. |
| REQ-SAFE-10 | Add an LLM-based approval reviewer, ensemble security analysers, or named policy rails alongside deterministic rules. Such reviewers supplement,... | may | Meets | Agent-based reviewers supplement the standing rules exactly as envisaged: the Ethics Council and Ground Truth Gatekeeper review independently, return pass/conditional/block, and may not waive the charter floor (skills/ethical-specialist-agent-network/SKILL.md). |
| REQ-SAFE-11 | Be available at Maximal conformance: per-turn or per-step snapshots with rewind and file restoration, so an approved-but-wrong action can be undone. | should | Partially meets | Artifact-level rollback is designed in — immutable baselines, superseding versions, rollback instructions required on build increments (skills/vibe-engineering-collaboration/SKILL.md; ARCH §8 'support rollback') — but per-turn filesystem snapshots with rewind are not specified. |

### 4.6 Multi-agent orchestration (ORCH) — 9 requirements: 3 Exceeds, 0 Meets, 4 Partially, 1 Does not meet, 1 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-ORCH-01 | Remain single-agent until a concrete breadth-first exploration phase is documented in which parallel, context-isolated workers clearly beat serial... | shall | Partially meets | The Lab is multi-agent by design, justified by staged specialisation and independent review rather than a documented breadth-first phase; it does carry the paper's restraint in local form — 'do not spawn when the parent can complete a small task coherently', smallest adequate team, sequential fallback (agents/README.md; skills/ethical-specialist-agent-network/SKILL.md). A deliberate divergence, recorded in the spec's Appendix A. |
| REQ-ORCH-02 | Where sub-agents exist, the pattern shall be coordinator–worker: a parent spawns workers, each with isolated (forked, linked, or fresh) context, a... | shall | Exceeds | Coordinator–worker is specified beyond the paper's sketch: the Director coordinates, every worker gets a bounded spawn contract and a typed packet result channel (MissionStagePacket/v1 and the domain packets), context is carried by packets — 'never rely on hidden conversational state' (agents/mars-harness-director/INSTRUCTIONS.md) — and the parent remains accountable for integration. |
| REQ-ORCH-03 | Spawn depth, concurrent-worker count, and per-worker turn/cost budgets shall be bounded and enforced. Unbounded recursive composition does not... | shall | Exceeds | Bounds are concrete and multiple: default maximum depth 3, child-count and iteration maxima in every spawn contract, per-child budget and timebox, PM repair caps of 20/50 cycles, stop at budget/time/evidence/acceptance/safety limits, and mandatory retirement with access revocation (agents/ethical-specialist-network-governor/INSTRUCTIONS.md). |
| REQ-ORCH-04 | A child agent’s permissions shall not exceed its parent’s: child capabilities shall be derived from, and intersected with, the parent’s permission... | shall | Exceeds | Permission non-expansion is a stated suite invariant: 'Permissions can narrow during delegation but never expand silently' (agents/README.md); a child 'receives no permission, data, budget, or tool access beyond the parent contract and cannot alter its limits', intersected with the ethics charter floor (skills/ethical-specialist-agent-network/SKILL.md). |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-ORCH-05 | Use in-process primitives (function calls, in-process sessions, thread forks). Cross-process coordination, where unavoidable, should use the... | should | Partially meets | Single-runtime coordination conforms — roles run in-process or sequentially with explicit role labels when spawning is unavailable (agents/README.md) — but cross-application coordination uses packet import/export rather than in-process primitives, a deliberate simplest-substrate choice specified as a contract, not a running bus. |
| REQ-ORCH-06 | Where many similar workers are needed, the harness should support declarative fan-out (for example a task list or CSV-driven spawn) with... | should | Partially meets | Declarative selection exists as rosters — standing specialist lists per director and the 34-role discipline bench (skills/docs/ENGINEERING_DISCIPLINE_AGENT_CATALOG.md) chosen per scope — but task-list/CSV-driven fan-out as a mechanism is not specified. |
| REQ-ORCH-07 | Support parallel delegation over conversation trees, sequential sub-agent delegation within a turn, swarm coordination over a shared store, or a... | may | Unverified | Optional Maximal patterns (conversation-tree parallelism, swarm over a shared store, mixture-of-agents) are not adopted as such; wiki memory is a shared store but no swarm coordination is claimed. Unverified. |
| REQ-ORCH-08 | Be measured per run and reported against the single-agent baseline for the same task class, so the roughly 15× cost risk | shall | Does not meet | Budgets are enforced per packet, but multi-agent token/cost consumption is never required to be measured per run or reported against a single-agent baseline; the ~15× cost risk has no visibility mechanism. No evidence in definitions. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | named in the evidence is... | | | |
| REQ-ORCH-09 | Be defined as files (Markdown with YAML frontmatter declaring name, model, tools, skills, hooks, and per-agent permissions) and resolved through a... | may | Partially meets | Agents are file-defined and registry-resolved — 11 agent folders resolved through agents/Mars_Harness_Agent_Manifest.json — but the files carry no YAML frontmatter declaring name, model, tools, skills, hooks and per-agent permissions; permissions sit in INSTRUCTIONS prose and spawn contracts. (Skills, by contrast, do carry frontmatter.) |

### 4.7 Extension surfaces (EXT) — 9 requirements: 1 Exceeds, 1 Meets, 5 Partially, 1 Does not meet, 1 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-EXT-01 | Provide at least one declarative extension surface beyond editing source code. At Minimal conformance this may be as small as structural typing /... | shall | Meets | Multiple declarative surfaces exist beyond any source code: skills, agent definitions, spawn and work-product contracts, SOUL/INSTRUCTIONS context files and platform adapters — the definitions are the system. Hooks, required only at Maximal conformance, are the missing surface (see REQ-EXT-05). |
| REQ-EXT-02 | Implement SKILL.md skills: directories with YAML frontmatter and Markdown instructions, discovered from project and user skill directories, usable... | should | Exceeds | SKILL.md is the Lab's first-priority mechanism, fully realised: 11 skill directories with YAML frontmatter (name, description), bundled docs, examples and contracts, and a test suite (skills/tests/validate_suite.py) that enforces frontmatter, required sections and phase minimums — validation the paper's corpus does not describe. |
| REQ-EXT-03 | Skill loading should be deferred: only catalogue metadata (name, description) is resident in the prompt; full instructions load on activation.... | should | Partially meets | Loading is staged in effect — the install model binds one skill per agent and the orchestrator routes stage → skill, so full instructions are not all resident at once, with README/manifest as catalogue — but metadata-only residency, on-activation loading and conditional activation by path/environment are runtime behaviours the definitions do not specify. |
| REQ-EXT-04 | Implement MCP for external integrations — external processes, services, databases, and internal APIs — as the second-priority | should | Partially meets | MCP is neither implemented nor mentioned. The conforming alternative (a documented minimal position) is only half-met: PLATFORM_ADAPTERS.md records a deliberate skills-plus-adapter stance, but no file records 'no MCP' as an explicit decision with rationale. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | extension mechanism.... | | | |
| REQ-EXT-05 | Expose lifecycle hooks (at minimum pre-tool, post-tool, and stop) through which extensions can observe, deny, rewrite inputs for, or append... | should | Does not meet | No lifecycle hook surface (pre-tool/post-tool/stop) exists through which extensions could observe, deny or rewrite a call; the Lab's gates are agent roles, not hook handlers. No evidence in definitions. |
| REQ-EXT-06 | Be bundled as plugins that contribute tools, skills, hooks, MCP configuration, agent definitions, and commands as one installable unit,... | may | Unverified | Optional: the suite is distributed as a licensed download package (LICENSE.txt, NOTICE.txt) but not as installable plugin units from a marketplace or registry. Unverified. |
| REQ-EXT-07 | Be treated as software packages: installation shall include trust-tier assignment, and at Maximal conformance shall include pre-install static... | shall | Partially meets | The Lab's own skills carry provenance (Apache-2.0 headers, version 5.1, license URL in every file) and structural validation; third-party installation controls — trust tiers, pre-install scanning, quarantine, provenance verification, path-traversal safety, dependency surfacing — are not specified. |
| REQ-EXT-08 | Support agent-authored skills: an agent creates or patches skills from completed work. Where enabled, new or changed skills shall pass through the... | may | Partially meets | Agent-authored improvement is envisioned only at roadmap level: Maturity Model level 5, 'Adaptive — post-run outcomes update playbooks through approved, versioned changes' (skills/docs/MATURITY_MODEL.md) — a human-approved curation path without a specified inbox or audit workflow. |
| REQ-EXT-09 | Configuration-driven extensibility (agent profiles, environment and runtime overrides, per-profile tool and permission sets) should be supported... | should | Partially meets | Behaviour varies without code changes through task packets (risk tier, permitted actions, budget, approval gates), the agent template, spawn contracts and platform adapters; a formal profile/override configuration system is thin — the only YAML configs are two skills/*/agents/openai.yaml interface files. |

### 4.8 Interoperability (INT) — 11 requirements: 0 Exceeds, 1 Meets, 6 Partially, 1 Does not meet, 3 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-INT-01 | Ship an ACP server, exposing its agent to editors over the | should | Does not meet | No ACP server is offered or mentioned; the Lab's interoperability bet is typed packets between its own applications. No evidence in definitions. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | protocol (including, where supported, session fork, workspace trust, and rewind). An… | | | |
| REQ-INT-02 | Consume other harnesses over ACP (harness hosting): running a peer harness as an interchangeable step-backend, or consuming an ACP agent as a… | may | Unverified | Optional: consuming peer harnesses over ACP is not described. Unverified. |
| REQ-INT-03 | Implement A2A for cross-vendor mesh topologies. A2A adoption shall be recorded as an explicit bet: in the corpus it is a single system's position… | may | Unverified | Optional: A2A is not adopted and no explicit bet is recorded. Unverified. |
| REQ-INT-04 | Be used for a harness's own internal sub-agent coordination where in-process primitives suffice (see REQ-ORCH-05). Standard protocols are for… | shall | Meets | Conformed to by design: neither ACP nor A2A is used for internal coordination — internal coordination is in-process role execution or typed packets (skills/mars-harness-orchestrator/SKILL.md). |
| REQ-INT-05 | Use the SKILL.md bundle format, and discovery shall check the conventional locations of neighbouring ecosystems as well as the harness's own… | shall | Partially meets | Skills use the SKILL.md bundle format with frontmatter and are designed to load across Codex, Claude Cowork, Hermes and Grok without rewriting (PLATFORM_ADAPTERS.md) — the stated portability goal — but discovery of neighbouring ecosystems' conventional skill locations is not specified, and cross-harness loading is asserted, not tested (validate_suite.py tests internal structure only). |
| REQ-INT-06 | Where the harness imports another harness's sessions, settings, context files, or | shall | Partially meets | A partial importer exists for the Lab's own applications: the Mission Memory Steward's application identity map namespaces native IDs from all three apps and appends crosswalks rather than collapsing them (skills/mission-memory-steward/SKILL.md). Importing other harnesses' |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | plugin formats, the importer shall target documented or... | | | sessions/settings, and the never-drop-safety-settings rule, are not addressed. |
| REQ-INT-07 | Be published and versioned by their owning harness. A harness adopting another harness's hook event names for compatibility should document the... | should | Unverified | No hook vocabulary exists to publish or map; the Lab's gate vocabulary (approved/conditional/blocked) is documented but not versioned as an interoperability surface. Unverified. |
| REQ-INT-08 | Skills and plugins distributed through a registry or marketplace should carry version, provenance, and trust metadata, with a versioned index that... | should | Partially meets | Distributed skills carry version (suite 5.1) and provenance (copyright/license/full-text URL in every file), but distribution is a direct download — no registry, versioned index or per-skill trust metadata exists. |
| REQ-INT-09 | Be offered as an importable SDK, and may expose the agent as a model behind an OpenAI-compatible endpoint, and may run as a client/server system... | may | Partially meets | The platform form is partly realised: three client/server SaaS applications implement the harness patterns and are offered hosted (spec Appendix A; agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md), but there is no importable SDK, no agent-as-model endpoint, and the cross-app embedded-API form is explicitly an unimplemented contract (ARCH §3). |
| REQ-INT-10 | Orchestrate multiple conforming harnesses behind one API. Where it does, it shall re-apply the expensive shared concerns at its own layer —... | may | Partially meets | The orchestrator/Director layer behaves as a meta-harness in design — routing three application harnesses behind typed packets and re-applying the expensive shared concerns (Ground Truth, ethics, human authority, memory) at its own layer, with a per-artifact route trace — but it is an architectural contract, not a running meta-harness (ARCH §3). |
| REQ-INT-11 | Be supported with governance surfaces where claimed: managed (for | shall | Partially meets | Governance mechanisms are named for the Lab's own tier — named Human Mission Authority, decision rights in PM charters, exportable per-agent audit packets — but managed (MDM-delivered) configuration and centrally administered policy/trust settings are not offered. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | example MDM-delivered) configuration, centrally administered policy and trust… | | | |

### 4.9 Non-functional requirements (NFR) — 14 requirements: 1 Exceeds, 3 Meets, 7 Partially, 0 Does not meet, 3 Unverified

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| REQ-NFR-01 | Determinism and replayability: Given the same model responses, a run shall be replayable from its recorded trace (REQ-LOOP-09): the harness shall… | shall | Partially meets | State is reconstructable at packet granularity — versioned packets, supersession chains, revision log, route trace, MemoryTransaction/v1 — but replay of a run from a recorded trace including model responses and compaction events is not specified. |
| REQ-NFR-02 | Determinism of retrieval: All code retrieval and default memory recall shall be deterministic: the same query over the same tree or transcript… | shall | Partially meets | Memory retrieval by stable IDs and explicit relationships is deterministic in design, and no similarity-ranked retrieval sits in any default path; no file states the same-query/same-order guarantee for search-style retrieval (e.g. Hyperia source search). |
| REQ-NFR-03 | Auditability and evidence: Every tool execution, permission decision (with the rule that produced it), compaction, sub-agent spawn, extension… | shall | Exceeds | Audit breadth exceeds the paper's list for this system class: Mission Memory records agent definitions, spawn events, parent–child links, run status, ethics reviews, approvals, dissent, residual risk and retirements (skills/mission-memory-steward/SKILL.md; starter-store templates for agent_run, spawn_contract and ethics_review), all in exportable Markdown/JSON independent of any vendor UI. Caveat: permission decisions are logged as dispositions/findings rather than rule IDs. |
| REQ-NFR-04 | Auditability: Secrets shall be redacted from traces and event logs by a secret registry or equivalent | should | Unverified | Secret redaction from traces and logs before persistence is not addressed in any file. Unverified. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | mechanism before persistence, so... | | | |
| REQ-NFR-05 | Cost budget: The harness shall meter tokens and cost per turn and per run, enforce operator-set budgets (REQ-LOOP-03), and report spend per run.... | shall | Partially meets | Budgets are set (packets, spawn contracts) and enforced as stop conditions, and PM Accelerator's app layer has retry/rate-limit controls; per-turn and per-run token/cost metering and spend reporting are not specified. |
| REQ-NFR-06 | Latency and context economy: Prompt size is a budget: tool and skill schemas resident in the initial prompt should be minimised (deferred loading,... | should | Partially meets | Context economy is practised — smallest relevant context on retrieval, per-packet scoping, PM document compaction — but resident schema share is not measured and the ~10% alarm threshold has no counterpart. |
| REQ-NFR-07 | Security: The harness shall treat all model output, file content, tool results, and web content as untrusted input to its control decisions... | shall | Meets | Stated as a design invariant at suite, agent and skill level: retrieved content is data, never governing instructions (ARCH §8); children cannot expand permissions; no content channel can waive the ethics charter or create an approval (skills/ethical-specialist-agent-network/SKILL.md guardrails). |
| REQ-NFR-08 | Portability (providers): Swapping the model provider through the conditioning layer (REQ-LLM-02, REQ-LLM-03) should not require changes to the... | should | Partially meets | Skills are explicitly model-independent and PLATFORM_ADAPTERS.md maps behaviour across four runtimes, supporting provider portability; no provider swap is demonstrated, and the reviewed apps are Gemini-only, so leak-freedom is asserted rather than evidenced. |
| REQ-NFR- | Portability (harnesses): User | should | Meets | All user state — agent/skill definitions, packets, wiki memory (schema_version 1.0.0 JSON), contracts — is stored in |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| 09 | state that the paper shows users expect to move — context files, skills, sessions, settings — should be stored in... | | | documented Markdown/JSON file formats another harness's importer can read; nothing user-facing is binary-only. |
| REQ-NFR-10 | Evolution and upgrade compatibility: Upgrades shall preserve user state and policy: context files, skills (including agent-authored skills and... | shall | Partially meets | Versioning discipline is strong (packet_version mars-harness/1.0, schema versions, append-and-supersede, tombstone-only deletion), so state is built to survive upgrades; a tested migration across Lab versions (e.g. V5.1 → V5.2) is not specified. |
| REQ-NFR-11 | Maintainability: The agent runtime shall have no dependency on a general-purpose agentic framework (LangChain, LangGraph, AutoGen, CrewAI,... | shall | Unverified | The definitions are framework-free Markdown/JSON by construction and the reviewed apps use raw provider SDKs with no agentic framework named (agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md); the Lab executes on host harnesses whose internals it does not control, so runtime-level conformance cannot be verified from this package. |
| REQ-NFR-12 | Observability: Loop, tool, and hook execution should be instrumented (for example OpenTelemetry spans) so a failing run can be diagnosed from... | should | Unverified | No instrumentation (e.g. OpenTelemetry spans) is specified; diagnosis rests on packets, run records and route traces. Unverified. |
| REQ-NFR-13 | Robustness: Malformed model output, oversized tool results, context overflow, and hosted-harness failure shall | should | Partially meets | ARCH §8 maps failure modes to recorded, recoverable states — blocked packets, branch isolation, versioned rollback, quarantine of corrupt inputs (Hypatia phase 6) — and retry is bounded where specified (20/50 PM cycles, 25 debugger attempts); context-overflow degradation is not addressed. |

| Req ID | Requirement (short form) | Level | Grade | Evidence note |
|---|---|---|---|---|
| | degrade into recorded, recoverable… | | | |
| REQ-NFR-14 | Scalability honesty: Claims about scale (tool counts, model counts, session counts) in conformance statements shall be dated inventory claims, in… | shall | Meets | Claims are dated and versioned throughout: package V5.1 dated 23 September 2026, manifest release date, source review dated 2026-09-23 against named branches, and explicit scoping of what is not claimed (ARCH §3). This assessment likewise states its basis (V5.1 package, assessed October 2026). |

## 5. Method and limitations

- **What was graded.** The V5.1 package as supplied: 11 agent definitions (INSTRUCTIONS.md + SOUL.md each), 11 SKILL.md skills, the agent manifest, the three-domain architecture, work-product contracts, maturity model, platform adapters, engineering-discipline catalog, ethical-specialist architecture, two stage-packet/memory examples, and the suite validation test — 51 files. Every one of the 97 requirements in the specification was graded individually; none was skipped or merged.
- **Design-level, not runtime.** The package is agent/skill *definitions and documentation*, not a running system. Grades therefore assess definition/design-level conformance: whether the behaviour is specified, not whether it has been observed executing. Runtime behaviour is unverified from these sources, and no grade in this document should be read as a test result.
- **Applications.** Where a grade rests on the three SaaS applications (Hypatia Pro, Intelligent Engineer Pro, PM Accelerator), it rests on the package's own source review (agents/docs/GITHUB_APPLICATION_SOURCE_REVIEW.md, 23 September 2026, `backup-2026-09-22` branches), not on fresh code inspection for this assessment. The package itself warns that harness controls around those applications 'must not be represented as native application behavior unless implemented and tested' — this assessment follows that warning.
- **V5.2.** The Lab's live site is at V5.2; the audited package is V5.1. Any V5.2 differences are unverified and could move individual grades in either direction.
- **Grade conventions.** *Exceeds* = meets the requirement and goes materially beyond the paper's model in that requirement's scope. *Does not meet* = the requirement demands a mechanism and the package — which presents itself as a complete design — contains no trace of it, or states a contrary position. *Unverified* = the requirement concerns host-runtime or platform behaviour the definitions neither specify nor contradict, or an optional (may) provision whose adoption cannot be determined. Each Unverified and Does not meet note says which case applies.
- **Traceability.** Requirement short forms are compressed from the specification's text; levels (shall/should/may) are the specification's own. Evidence notes cite the Lab file that carries the claim, or state that no Lab source addresses the requirement.